Learning with less: label-efficient land cover classification at very high spatial resolution using semantic stratification and self-supervised deep learning

Author order?

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Abstract

Deep learning methods for land cover classification have been shown to be highly performant, particularly at high spatial resolutions where traditional pixel-wise or object-based approaches struggle. However, the wide parameter space of deep neural networks typically requires several thousand densely annotated imagery tiles to achieve reliable performance. This is problematic, as data annotation is costly and time-consuming, creating a strategic barrier to widespread adoption of such models for land cover classification at high resolutions. We address this challenge through two complementary approaches. First, we demonstrate that utilizing a stratification approach that groups candidate imagery tiles into semantic clusters using existing moderate resolution land cover products improves the quality of a training dataset compared to simple random sampling. Furthermore, applying Bootstrap Your Own Latent (BYOL) self-supervised learning to an ImageNet pre-trained ResNet-101 encoder using unannotated aerial imagery enhances the performance of a semantic segmentation network on a downstream land cover classification task, particularly when few samples are supplied during training. When applied to a 1 m resolution land cover task using linear probing, our BYOL model demonstrates average increases of 13.04% and 14.87% in mIoU and overall accuracy over a similar model pre-trained on ImageNet alone, illustrating the extent of the domain gap between general-purpose imagery often used for pre-training and high resolution remote sensing data. Using only 500 labeled samples during the training phase, our approach achieves 87.42% overall accuracy and 63.00% mIoU for comprehensive 8-class land cover mapping across Mississippi when the model is fine-tuned in a U-Net framework—improvements of 2.06% and 2.77% over ImageNet pre-training. These results demonstrate that creative approaches to sample stratification and selection along with advanced pre-training can substantially reduce the need for hand-annotated data, breaking down hurdles for large-scale high spatial resolution land cover mapping.

1 Introduction

Land cover data provides valuable information about the characteristics of the Earth's surface for a wide variety of applications across many domains, including agriculture, urban planning, and environmental monitoring Ban et al. (2015); Loveland et al. (2000); Scarpace and Quirk (1980); Townshend (1992); Wulder et al. (2018). Research into automated methods for land cover classification from remotely sensed data has been ongoing for several decades, with a majority of methods relying on supervised learning techniques to classify land cover using multispectral aerial or satellite imagery Maxwell et al. (2018); Phiri and Morgenroth (2017). Historically, pixel-wise classification methods have been the predominant approach for land cover classification, where each pixel in the image is classified based on the pixel's raw spectral values, hand-crafted features, and ancillary data from other geospatial datasets Khatami et al. (2016); McIver and Friedl (2001). This pixel-wise paradigm is effective for classifying coarse and moderate spatial resolution imagery (e.g., MODIS, Landsat, Sentinel-2), but is less effective for classifying high spatial resolution imagery due to decreased inter-class spectral separability and increased intra-class spectral variability Grekousis et al. (2015); Hansen and Loveland (2012); Lu and Weng (2007); Markham and Townshend (1981); Strahler et al. (1986); Woodcock and Strahler (1987). This is problematic, as ideally, high spatial resolution (< 10 m) land cover data offers more detailed information about the Earth's surface with greater spatial accuracy, which is valuable for applications that require fine-grained information about land cover types and their spatial distribution Fisher et al. (2018); Li et al. (2022a). Object-based Image Analysis (OBIA) techniques have been proposed as a potential workflow to address these issues, but their performance is highly dependent on the quality of objects or superpixels extracted from the imagery during the segmentation process, which must be carefully tuned to the specific characteristics of the imagery and the land cover classes of interest Blaschke et al. (2014); Johnson and Xie (2011); Ma et al. (2017). Still, a key idea behind OBIA is that the spatial information contained in the objects can be leveraged to provide more information to the classifier, which addresses the inter-class spectral separability issues that hamper pixel-wise classification methods Blaschke et al. (2000); Myint et al. (2011); Tehrany et al. (2014). The application of deep computer vision models to land cover classification has been of significant interest in recent years, as these models excel at extracting spatial and spectral features from imagery in a data-driven manner, making them well-suited for high spatial resolution land cover classification tasks Ma et al. (2019); Vali et al. (2020); Zhu et al. (2017).

Convolutional neural networks (CNNs) are a class of deep learning models that have been widely used for land cover classification tasks due to their ability to automatically extract hierarchical spatial structures from the input imagery that are useful for classifying land cover type Kussul et al. (2017); Li et al. (2022b); Pan et al. (2020); Song et al. (2019). CNN-based semantic segmentation models learn a mapping from the input image to a pixel-wise classification probability map, which can be used to generate a land cover map by assigning each pixel to the class with the highest probability Long et al. (2015). Semantic segmentation models have been widely used for land cover classification tasks, as they can facilitate the generation of high spatial resolution land cover maps that leverage both spectral and spatial information in the input imagery under a unified framework Yuan et al. (2021). However, these models require large amounts of labeled training data to learn the complex spatial and spectral patterns present in the imagery, which can be difficult to obtain for high spatial resolution imagery due to the high cost of collecting and labeling the data Hu et al. (2015).

This need for large amounts of ground truth data for accurate training of deep learning models has led to the extensive research into methods for reducing the amount of data necessary to train these models. Most of these methods focus on leveraging transfer learning, where the parameters of a model pre-trained using a large dataset are transferred to a model that is subsequently fine-tuned on a smaller dataset of interest. Transfer learning is effective for deep computer vision models, as the features learned by a model on a large dataset are often generalizable to other datasets thanks to the hierarchical nature of the features learned by the model Hu et al. (2015); Kolesnikov et al. (2020). However, the effectiveness of transfer learning is highly dependent on the similarity between the source and target datasets, as well as the amount of labeled data available for the target dataset. For example, a common approach to transfer learning for land cover classification tasks is to pre-train a model on a large general-purpose image classification dataset (e.g., ImageNet Russakovsky et al. (2015)) and then fine-tune the model on a smaller dataset of interest Pires de Lima and Marfurt (2020). This is generally effective, but the domain shift between the source and target datasets can hinder the performance of the model, and the extent of this domain shift between general purpose image classification datasets and high spatial resolution remote sensing datasets is not well understood Ma et al. (2024). Self-supervised pre-training methods have been proposed as an alternative to traditional fully supervised pre-training methods, as they can leverage the spatial and spectral information present in the input imagery to learn useful features through the use of pretext tasks that do not require labeled data Chen et al. (2020); Tian et al. (2020); Wu et al. (2018). In remote sensing, these methods are becoming more widely explored as a means to pre-train large foundation models that can serve as strong feature extraction backbones for downstream tasks **CITE!**. While effective, foundation models are geared for performance in multi-sensor, mutli-modal, or cross-task scenarios, and not strictly label efficiency **CITE!**. As such, to the authors' knowledge, little work has been done on how these self-supervised pre-training approaches can impact operational land cover mapping under extreme low-data scenarios.

Beyond the choice of pre-training routine, another poorly understood component of developing semantic segmentation models for land cover classification is the choice of sampling routine used to select training data. Ideally, a simple random sample of sufficient size from the region of interest should produce a quality training dataset. However, the sample size must be large enough to satisfy the conditions that the sampled data is representative of the variation in land cover across the region while also having enough samples in the minority classes such that a classifier can capture said classes accurately. Simple random sampling becomes problematic for large areas with diverse land cover types or strongly imbalanced land cover distributions. Much research on stratification strategies for building pixel-based datasets for land cover classification has been conuducted to address the issues associated with simple random sampling. These stratification procedures involve splitting the population into several groups based on some intrinsic property (e.g., geospatial location) or information from an external data source (e.g., prior land cover information) and sampling from these groups proportionally. However, semantic segmentation models require small patches consisting of dense labels, that is, every pixel in the patch is assigned to a class, rather than individual pixels. While this does not negate the implications of stratification, it does add extra considerations to the process, as one patch can have multiple classes. For example, when stratifying training data by land cover class according to some prior knowledge of the land cover distribution, it can be very difficult to select patches in such a way that fit the desired land cover distribution. Further, the composition of classes in sampled patches should also be representative of spatial relationships between land cover types in the region of interest. **SEE** `https://chatgpt.com/c/6894f587-a918-832a-8112-73ab5926bb09`, `https://www.mdpi.com/2072-4292/15/6/1607#:~:text=Figure%205C%2C%20we%20show%20that,the%20reported%20confusion%20matrix%20accuracies` This issue is exacerbated under low-data scenarios, where it can be difficult to capture the full diversity of both land cover types and their spatial relationships with few samples.

In this work, we demonstrate that employing a novel stratification strategy that groups candidate patches according to their semantic content (i.e., land cover composition) using a coarse-resolution land cover dataset to guide the stratification selection such to sample patches that are representative of the diversity of land cover compositions found in the region of interest. Additionally, we employ Bootstrap Your Own Latent (BYOL) as a self-supervised pre-text task during pre-training, enabling a ResNet-101 convolutional encoder network to learn to extract deep features from aerial imagery without the need for full supervision using labelled data. This approach is used to tackle land cover classification at 1 m resolution over the state of Mississippi using only 1,000 patches selected using the semantic stratification procedure. We demonstrate that these approaches can be used to train deep semantic segmentation models that are accurate, generalizable, and capable of yielding acceptable performance even when supplied with only 250 patches of imagery-ground truth pairs during the supervised fine-tuning stage.

2 Related work

2.1 Approaches to very high spatial resolution land cover classification

maybe define VHSR in the intro???

VHSR land cover classification has historically been tricky due to blah blah blah Prior to deep learning, OBIA was the predominant paradigm - mention a few models Performance of OBIA at VHSR attributable to integration of spatial information into imagery classification Drawbacks, scale parameters, generalization, human intervention

Semantic segmentation introduced in computer vision community Why it worked well Prominence in VHSR classification today List a couple of notable works on convolution

2.2 Training data sampling strategies

Mention stratification, classical pixel-wise Struggles when applying to patch wise Mention a few

2.3 Remote sensing pre-training and self-supervised learning

Bring up pre-training, why its important, how transfer learning works Difficulty in remote sensing pre-training, unsure of domain shift betweem standard ImageNet pre-training and remote sensing imagery Emergence of self-supervised learning as a new paradigm for pre-training

How this has translated to remote sensing in the form of foundation models What these foundation models do, why they are built Problems with foundation models: ViT lacks inductive bias toward imagery Transformers still not the norm in remote sensing analysis, lack of inductive bias makes very low data fine-tuning difficult Still, can apply SSL to CNNs using contrastive/self-distillation approaches

3 Study area

Mississippi, USA The overarching goal of this proposed work is to develop a 1 m resolution land cover map for the state of Mississippi to facilitate high-fidelity land analysis. Mississippi is located in the southeastern United States, and is home to a diverse range of land cover types. The region's humid subtropical climate and fertile soils in the Mississippi River Delta make these areas home to a variety of agricultural activities Mississippi State University Extension, while the remainder of the state is characterized by forests and woodlands which comprise the majority of land cover in the state United States Geological Survey. The coastal region along the Gulf of Mexico is home to a sprawling urban network intertwined with herbaceous wetlands and barren beaches English (2011). The state's diverse land cover types, geographical position along the Mississippi River and Gulf of Mexico, and economic dependence on agriculture and natural resources make the need for accurate land cover information at high spatial resolution a priority in order to support natural resource management, agricultural planning, and environmental monitoring efforts.

Virginia, USA In order to validate our proposed semantic stratification approach, we also invesrtigate the performance of the proposed method using an existing land cover dataset. The Chesapeake Bay Program (CBP) land cover dataset is used to provide ground truth data for this phase of the study. The CBP land use and land cover products are among the largest meter-scale land cover datasets in the United States, and are produced by the Chesapeake Conservancy as part of a cooparative agreement between the US Geological Survey and the US Environmental Protection Agency Claggett et al. (2025). The 2024 edition provides a 56-class land use product at 1 m resolution over the Chesapeake Bay watershed, which includes portions of six states (Delaware, Maryland, New York, Pennsylvania, Virginia, and West Virginia) and the District of Columbia for the years 2013/2014, 2017/2018, and 2021/2022.

National Agriculture Imagery Program (NAIP) For this study, USDA National Agriculture Imagery Program (NAIP) imagery is used as the primary data source from which the models are trained and the land cover map is generated. NAIP imagery is high-resolution (0.6 meter) multispectral aerial imagery collected during the summer months over the contiguous United States. Such imagery is collected on a 2–3 year cycle (depending on the state) and is predominantly used for agricultural purposes. While the temporal availability of NAIP imagery is limited, the high spatial resolution of the imagery and public availability of the data make it an attractive data source for remote sensing applications that require high spatial resolution data over large areas in the contiguous United States Earth Resources Observation and Science (EROS) Center (2017). NAIP imagery collected in 2021 and 2023 for the states of Virginia and Mississippi, respectively, is used in this study. The imagery is provided in 4 spectral bands (Red, Green, Blue, and Near-Infrared) at a spatial resolution of 0.6 meters per pixel. The imagery is resampled to a spatial resolution of 1 m to reduce the computational cost of processing and storing the data, as well as to align with the desired spatial resolution of the land cover product we aim to generate. Mississippi imagery is reprojected to the Mississippi Transverse Mercator (MSTM) coordinate system (EPSG: 3813) in accordance with recommendations from the Mississippi Automated Resource Information System (MARIS) to ensure that all data is in the same coordinate reference system Misssissippi Automated Resource Information System. Likewise, data for the Chesapeake Bay program is reprojected to CONUS Figure 1 shows the extent of the study area and the NAIP imagery used in this study in the MSTM coordinate system.

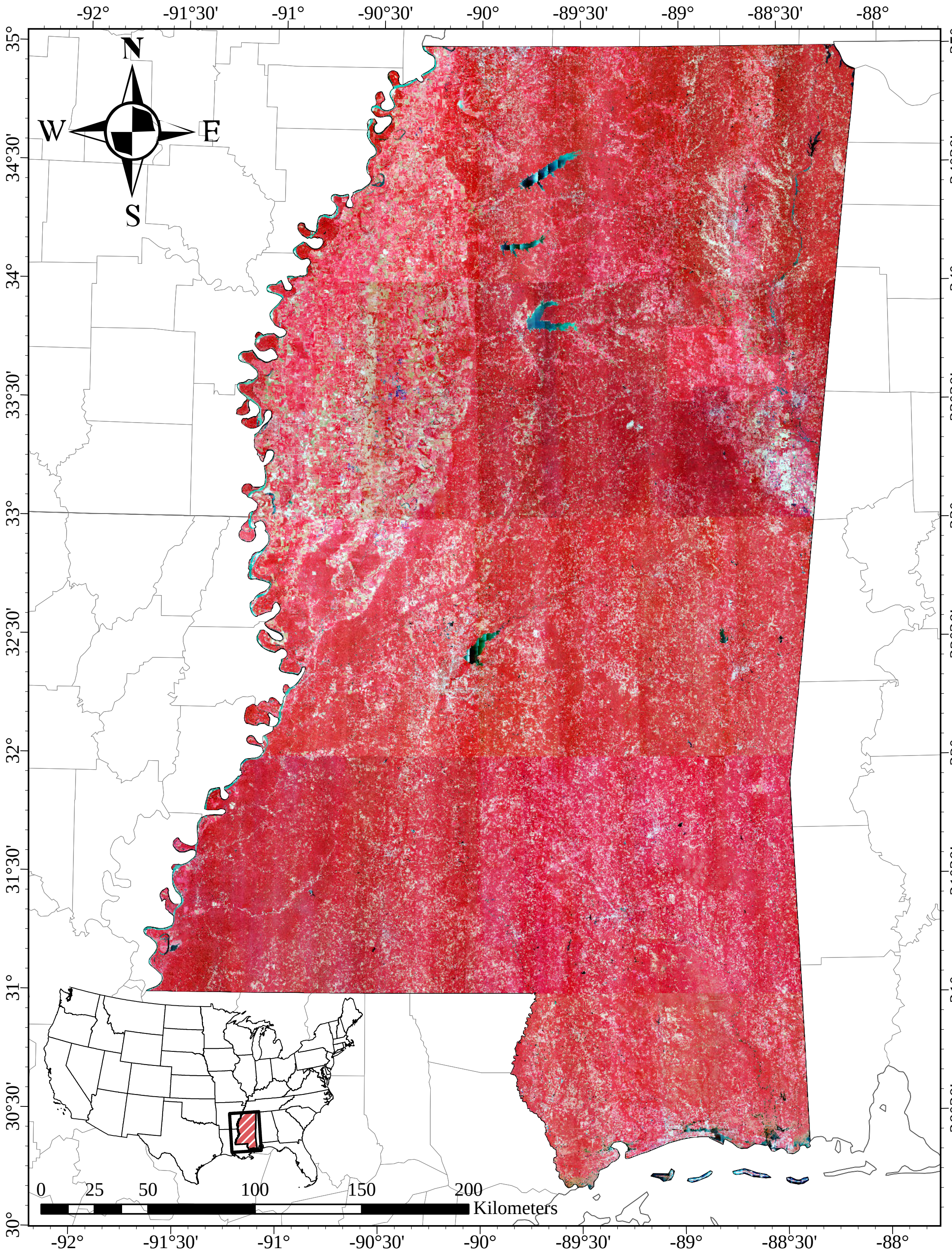

Figure 1: 2023 NAIP color-infrared imagery over Mississippi, USA.

4 Methodology

Our proposed methodology consisted of a multi-stage approach to land cover classification that aimed to produce a land cover product at high accuracy under data-scarce conditions using a machine-learning driven stratification protocol that aims to extract informative samples for training and the use of self-supervised learning to pre-train a large vision model for feature extraction from color-infrared aerial imagery in the absence of ground truth data. First, we applied the semantic stratification approach to train and evaluate a land cover classification model using a subset of the CPB land use/land cover dataset in Virginia in order to validate the approach in the context of other stratification approaches. We then used this approach to select 1000 sample tiles over the state of Mississippi for manual visual annotation to build a training dataset. A separate, larger dataset of unannotated sample tiles from the state of Mississippi was used to pre-train a ResNet-101 convolutional encoder using BYOL as a pre-text task under a self-supervised learning framework. The ResNet-101 encoder was then used as a backbone for a semantic segmentation architecture that was fine-tuned using the annotated training dataset. Finally, several fine-tuned models were ensembled to produce a land cover map at 1 m resolution over the state of Mississippi for the years 2016 and 2023, with the 2023 map undergoing a thematic accuracy assessment using a point-based validation approach.

4.1 Validation of semantic stratification approach using CPB land use/land cover data

Our proposed semantic stratification approach relied on the use of existing land cover data to form strata according to the distribution and composition of land cover classes within a population of potential training samples over a given area. Given an area of interest, we overlaid a grid of 256×256 meter cells that intersected with the study area, and these cells served as the population of potential training samples. We then extracted zonal histograms of land cover class distribution within each cell using the NLCD land cover data as a reference. The NLCD classes were reclassified down to 7 classes in accordance with the chosen land cover classification scheme, as shown in Table 1. The zonal histograms served as an informative feature vector for each cell, which effectively described the land cover composition within each cell. We scaled these histograms using the median and interquartile range of the histograms across all cells in the study area to ensure that the features were robustly normalized; rank exploratory statistics were chosen over mean and standard deviation statistics to avoid the influence of outliers. Principal component analysis (PCA) was then applied to the scaled histograms to reduce the dimensionality of the feature space and to extract the most informative features. We selected 2 principal components to represent the feature space of the land cover composition vectors. K-means clustering was then applied to the PCA-reduced feature space to form 250 strata, which were then used to stratify the population of potential training samples. A large number of clusters was chosen to ensure the strata were sufficiently fine-grained to capture the variability in land cover composition across the study area. From each stratum, we randomly sampled 4 candidate training samples, with each being placed into a separate fold of the training dataset such that 4 folds formed a total of 1000 training samples. The full stratification process is illustrated in Figure 2.

Table 1: Reclassification scheme for aligning NLCD land cover classes with our desired classification scheme.

NLCD Class Index	**NLCD Class**	**Target Class Index**	**Target Class**
11	Open Water	1	Open Water
23 24	Developed, Medium Intensity Developed, High Intensity	2	Impervious Structures
21 22	Developed, Open Space Developed, Low Intensity	3	Impervious Surfaces
31	Barren Land	4	Barren
41 42 43 90	Deciduous Forest Evergreen Forest Mixed Forest Woody Wetlands	5	Forest
52 71 81 95	Shrub/Scrub Grassland/Herbaceous Pasture/Hay Emergent Herbaceous Wetlands	6	Herbaceous
82	Cultivated Crops	7	Cultivated Crops
250	No Data	0	No Data

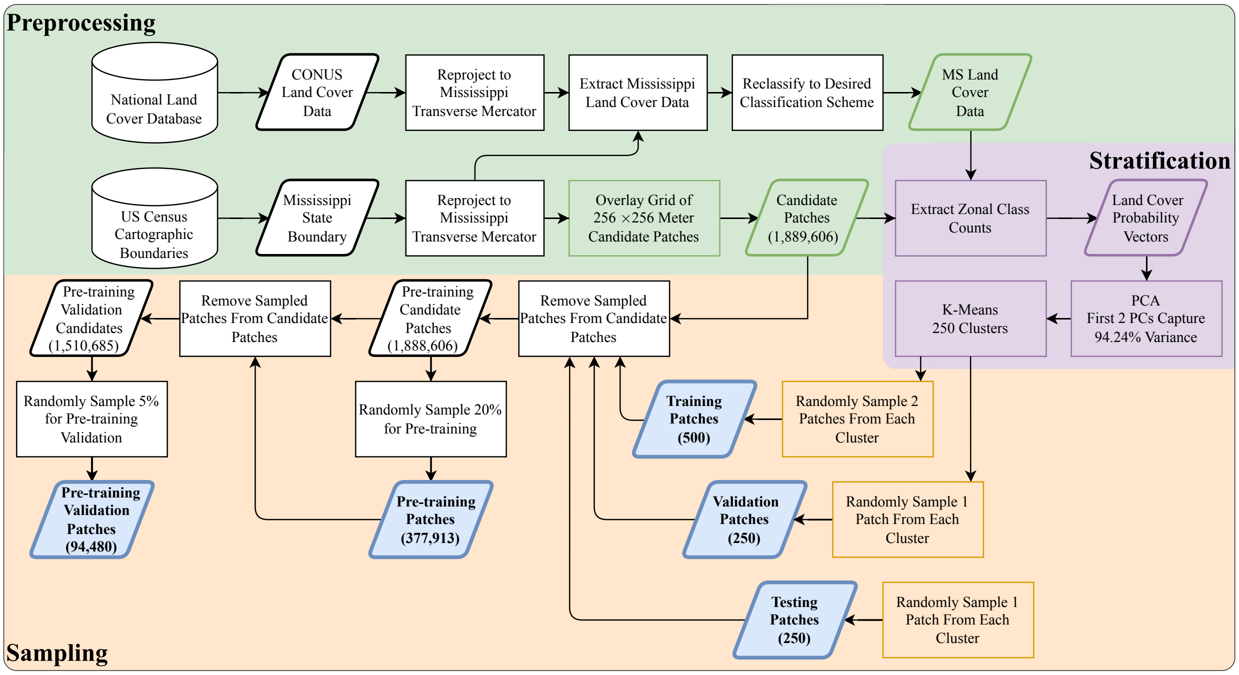

Figure 2: Flowchart of the stratified sampling process for selecting data for model training and validation **GO BACK AND FIX ME!!!**.

In order to validate the effectiveness of the proposed stratification approach, we applied it to the task of land cover classification using the CPB land use/land cover dataset. As comparisons, we also applied simple random sampling and spatially stratified sampling approaches to the same dataset. To form spatial strata, we treated the grid of candidate training samples as a graph where each cell is a node and edges were formed between neighboring cells. The METIS graph partitioning software was then applied to the graph to form 250 spatial strata that were spatially contiguous and had similar sizes using parallel multilevel graph partitioning algorithms (Karypis and Kumar, 1997, 1998; Kloeckner et al., 2025). As with the semantic stratification approach, we sampled 4 candidate training samples from each stratum, with each sample being placed into a separate fold of the training dataset such that 4 folds formed a total of 1000 training samples. We employed a stratified k-fold cross-validation approach to evaluate the performance of the land cover classification models trained using the three sampling approaches across 3 different potential training dataset sizes: 250, 500, and 750 samples. Figure 3 illustrates the division of the training dataset into folds for the three different training sample sizes.

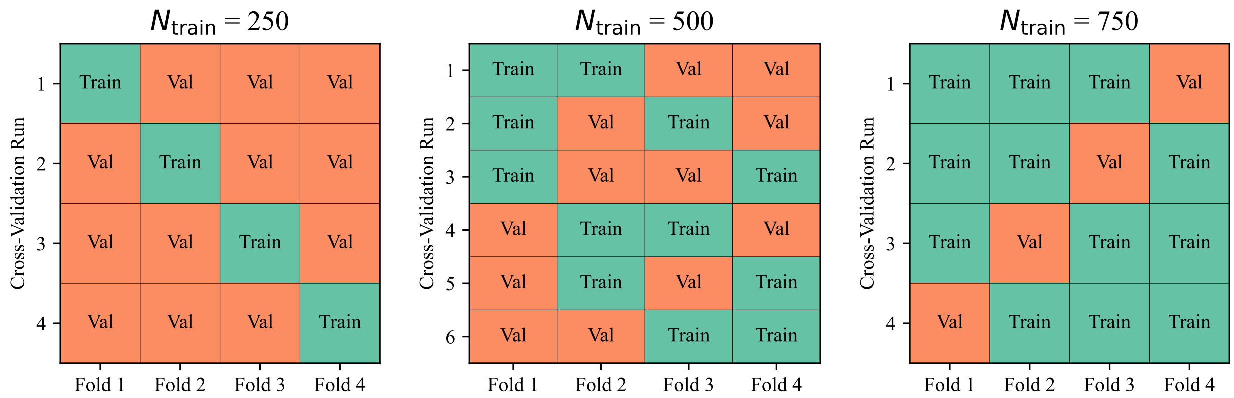

Figure 3: Folds of the training dataset used for k-fold cross-validation under 250, 500, and 750 training samples per fold.

For the CPB land use/land cover dataset, we produced and trained on 10 separate replications of simple random sampling, spatial stratification, and semantic stratification approaches to ensure that any observed differences in the performance of the models trained using these approaches were reproducible and not due to random chance. **The land use classes were reclassified down to 7 classes as shown in...** Further, 50,000 samples were randomly selected from the remaining candidate samples in the CPB land use/land cover dataset to form a large test dataset that was used to evaluate the performance of the trained models.

Candidate samples that lay within 500 meters of a sampled training samples were excluded from consideration for the test dataset to ensure that spatial autocorrelation did not lead to data leakage between the training and test datasets. Metrics were computed both on an overall basis and on a per-class basis to evaluate the performance of the models. Specifically, we chose to utilize the mean Intersection over Union (mIoU) metric as the primary evaluation metric, as it is a commonly used metric for semantic segmentation tasks and is robust to class imbalance as the IoU for each class contributes equally to the overall metric. mIoU is computed as shown in Equation 1, where C is the number of classes, TP_c is the number of true positives for class c, FP_c is the number of false positives for class c, and FN_c is the number of false negatives for class c.

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^{C} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{1}$$

A DeepLabv3+ architecture with a ResNet-101 encoder network served as the model for which the semantic stratification approach was validated. A separate model was trained for each run of the cross-validation framework, with the model being trained for a maxmimum of 1000 epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 32. Focal loss was used as the loss function to address the class imbalance in the land cover classes as shown in Equation 2, where p_t is the model's estimated probability for the true class amd $\gamma = 2.0$. A warmup period of 10 epochs was used to linearly increase the learning raate from 1×10^{-5} to 1×10^{-4} at epoch 10, after which the learning rate was decayed by a factor of 0.1 after the validation loss fails to decrease for 15 epochs. After 50 epochs of no improvement in the validation loss, training was halted and the model with the lowest validation loss was selected as the final model for that run. Automatic mixed precision training was used to speed up the training stage and reduce memory usage. Each model was evaluated on the test dataset using the mIoU metric, and the results were averaged across the 10 replications and k-folds for each training dataset size and sampling strategy.

$$\mathcal{L}_{\text{Focal}}(p_t) = -(1 - p_t)^{\gamma} \log(p_t) \tag{2}$$

4.2 Dataset curation and SAM-assisted visual annotation

Once the semantic stratification approach was validated using the CPB land use/land cover dataset, we applied it to the state of Mississippi to select 1000 candidate training samples for manual visual annotation. A team of trained annotators was employed to visually annotate the selected samples according to the land cover classification scheme shown in Table 2. Samples were randomly assigned to each annotator without regard to the sampling strategy used to select them such that each fold contained a mix of samples from each annotator. After the annotators completed their work, the annotations were reviewed and corrected by another annotator whose task was to ensure that the annotations were consistent and accurate. Annotators employed the Segment Anything Model 2 (SAM 2) hosted on the Computer Vision Annotation Tool (CVAT) annotation platform to assist with the visual annotation process CVAT.ai Corporation (2023); Ravi et al. (2024). By using positive and negative point prompts to create an initial mask corresponding to the boundaries of land cover classes in the imagery, annotators were able to quickly refine these masks to produce accurate annotations in a fraction of the time it would take to manually draw the masks.

Table 2: Land cover classification scheme for the Mississippi dataset.

Index	Class	Description
1	Open Water	Rivers, lakes, ponds, and other standing or moving water. Water bodies typically appear dark in CIR imagery due to near-infrared absorption; may appear bluish-green depending on depth and turbidity.
2	Impervious Structures	Elevated built features (e.g., buildings, sheds). Bright and reflective with sharp-edges and shadows.
3	Impervious Surfaces	Paved areas such as roads and parking lots. Typically bright but may appear dark depending on material. Linear patterns and lack of shadows distinguish from barren land and impervious structures.
4	Barren Land	Exposed soil, sand, or rock with minimal vegetation. Bright or neutral in CIR imagery, often interleaved with herbaceous areas.
5	Forest/Woody Vegetation	Tree-covered areas with dense canopy. Deep red in CIR imagery due to high near-infrared reflectance. Uneven texture with large shadows.
6	Herbaceous/Low Vegetation	Short vegetation, including grasses and shrubs. Smoother texture than forest, lighter red in CIR imagery. Common in clearings and field edges.
7	Cultivated Crops	Actively farmed fields, often in distinct rows. Bright red or pink in CIR imagery at peak growth. Presence of rows or patterns distinguish from herbaceous areas.
8	Unclassified	Shadows or occlusions obscuring surface features. Deep black regions near trees and buildings.

Aside from the samples selected for annotation, a large dataset of unannotated samples was also collected from the state of Mississippi using simple random sampling to form a large dataset of unannotated samples for pre-training the ResNet-101 encoder. 20% of the total number of candidate samples were selected for this dataset, while 5% of the candidate samples were selected to serve as a pre-trained validation dataset to monitor for overfitting during the pre-training process, corresponding to 377,921 and 94,480 samples, respectively. Once all sample tiles were selected, a dataset of 25,000 points was created by randomly sampling points from the remaining candidate samples to serve as a validation dataset for the fine-tuning of the semantic segmentation architecture and to facilitate a point-based thematic accuracy assessment of the final land cover map. No points within 200 m of a sampled tile were selected to prevent data leakage between the training and validation datasets caused by spatial autocorrelation, and no points within 1 km of a previously sampled point were selected to ensure that the validation points were spatially independent of each other.

4.3 Self-supervised pre-training using BYOL

The pre-training phase of the methodology is intended to leverage the large dataset of unannotated samples such that a ResNet-101 encoder can be trained to extract relevant features from the imagery that are well-suited for downstream land cover classification tasks. BYOL uses a "self-distillation" approach to learn representations from images using two separate but inter-dependent networks: an online network that conists of the primary encoder and a target network that consists of an exponential moving average (EMA) of the online network's parameters. Given an input image, two separate augmented views of the image are created using a set of transformations, as showni in . Each view is then passed through the online and target networks to produce feature vectors, as shown in Figure 4. The online network is trained to predict the feature vector produced by the target network given the other view as input, and backpropagation is used to update the weights of the online network based on the However, the target network is not trained directly; instead, it is updated using an EMA of the online network weights, giving the online network a stable yet constantly updated target to learn from. The online network consists of an image encoder, a multi-layer perceptron (MLP) projection head, and an MLP prediction head, while the target network consists of an image encoder and an MLP projection head. The online encoder's paramaters are updated with respect to the loss function in Equation 3, where p is the output of the online network and z is the output of the target network.

$$\mathcal{L}_{\text{BYOL}} = 2 - 2\frac{\langle p, z\rangle}{\|p\|\|z\|} \tag{3}$$

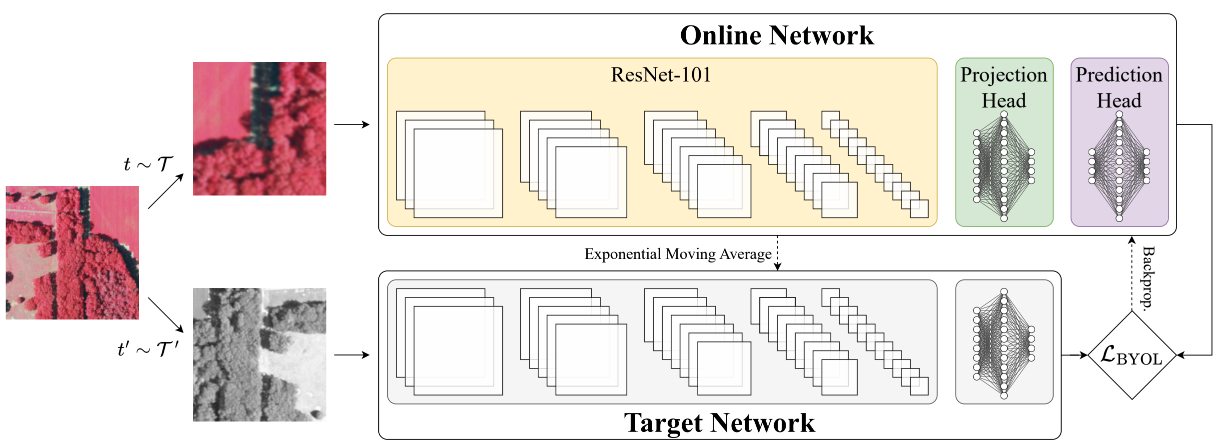

Figure 4: BYOL pre-training step.

The target network's parameters θ_{target} are updated according to the EMA update rule shown in Equation 4, where θ_{online} is the current set of parameters for the online network and τ is the decay factor for the EMA operation, and is increased from 0.996 to 1 over the course of pre-training according to a cosine annealing schedule.

$$\theta_{\text{target}} \leftarrow \tau\theta_{\text{target}} + (1-\tau)\theta_{\text{online}} \tag{4}$$

We applied BYOL to a ResNet-101 encoder network using the large dataset of unannotated samples from the state of Mississippi. We tested two configurations of the BYOL pre-training process: one starting with ImageNet pre-trained weights at the beginning of the pre-training process and one starting with randomly initialized weights. The pre-training process was run for 300 epochs, with the learning annealed from 1×10^{-3} to 0 over the course of training using a cosine annealing schedule, with a warmup period of 10 epochs to linearly increase the learning rate from 0 to 1×10^{-3}. The AdamW optimizer was used with a batch size of 4096: to achieve this large batch size while working withing the memory constraints of a single Nvidia A100 GPU with 80 GB of VRAM, we employed gradient accumulation using a microbatch size of 256.

4.4 Selecting a semantic segmentation architecture and fine-tuning

Table 3 lists the semantic segmentation architectures that were evaluated in this study.

Table 3: Semantic segmentation architectures evaluated in this study, with their total parameters using a ResNet-101 backbone.

Model	**Number of Parameters**	**Reference**
Attention U-Net	83,610,272	(Oktay et al., 2018)
BiSeNet	49,535,880	(Yu et al., 2018)
DANet	57,188,426	(Fu et al., 2019)
DeepLabV3+	58,199,128	(Chen et al., 2018)
PAN	62,439,496	(Li et al., 2018)
PSPNet	122,216,520	(Zhao et al., 2017)
UNet	92,302,024	(Ronneberger et al., 2015)
UPerNet	73,251,400	(Xiao et al., 2018)

4.5 Model ensembling and large-scale land cover mapping

5 Results

6 Discussion

7 Conclusion

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